

Under Irrigation Detection from Aerial Imagery

A technical write up for the WaqAla and Rizoma AI Engineer assessment, by Abdullah Bahri

The problem

Given an aerial photo of a field, I need to mark the areas that look under irrigated and give a percentage of the image they cover. A percentage only makes sense if every pixel is actually classified, since a dry patch is an irregular blob, not a box. That is why I built a segmentation model rather than a detector or a classifier.

The dataset

No public dataset labels under irrigated directly, so I looked for the closest honest proxy instead of guessing. Agriculture Vision, from Chiu et al., CVPR 2020, has nine expert annotated anomaly classes, and one of them, drydown, is defined as vegetation stress from insufficient water. Its licence allows non commercial academic and research use. Whether a paid assessment counts as academic use is genuinely unclear to me, so I asked WaqAla directly rather than assuming. I feed the model four channels, red, green, blue, and near infrared, since the original paper reports a real gain from adding infrared on this exact task.

The split and the baseline

Fields never appear in more than one split. My first attempt at splitting fields for validation and test still produced a real imbalance, 2,768 positive images in one half against only 1,725 in the other, even with equal field counts. I fixed it with a split that balances the actual share of positive images, which brought the two halves to 2,247 and 2,246. Before training anything I built a baseline with no learning at all, NDVI thresholded at whatever cutoff scored best on validation, and any trained model had to clear that bar to be worth reporting.

The model

I used DeepLabV3+ with a pretrained encoder, the same architecture the original paper used as its own strongest baseline for this class. The first pass, a ResNet 34 encoder on 800 images with no augmentation, overfit hard, training loss fell to about 0.02 while validation had already peaked by epoch sixteen. The second pass addressed that directly with a ResNet 50 encoder, 3,000 images instead of 800, real augmentation, and early stopping. I also built a third, exploratory approach using YOLO26, a model released in January 2026 with support for both semantic segmentation and multi channel input, trained on the identical split for a fair comparison.

Results

Each model's own held out test set, never touched during training or calibration:

Model	NDVI baseline IoU	Model test IoU	Improvement
First pass, ResNet 34	0.3355	0.5342	+0.1987
Second pass, ResNet 50	0.3373	0.5790	+0.2417
YOLO26, exploratory	n/a	0.5700	n/a

I also checked all three models against one shared pool of 2,000 images that none of them had seen in any role, not training, not validation, not even each model's own test set, a stricter check than any single model's own number (full detail in Models Comparison/Models_Comparison.ipynb):

Model	Accuracy on the unseen pool
First pass, ResNet 34	81.3%
Second pass, ResNet 50	88.1%
YOLO26, exploratory	85.65%

The second pass wins on both checks, so that is the model I am submitting. Threshold calibration mattered more than I expected too, the default cutoff of 0.5 understated the second pass model by about ten points of IoU against its real optimum of 0.45.

What confuses the model

The worst failure case was a completely clean field with 96 percent of its area flagged as dry. The image has an unusual purple and blue mottled texture, and the same texture shows up in the model's correct detections too, genuinely overlapping with real drydown there. The model has learned to treat that texture as its main signal for dryness, which works most of the time in this dataset but fails completely when a clean field happens to share it for an unrelated reason, most likely a difference in camera calibration between capture years, since the source paper documents exactly that. That is a real, explainable weakness, not a random error, and more of the same kind of data would not necessarily fix it.

Scope, Saudi conditions, and deliverables

A full hyperparameter search, covering all nine anomaly classes instead of just this one, and fine tuning a foundation model like SAM were all real options I set aside, since none of them would have paid for their own time cost inside an eight to ten hour budget. Agriculture Vision is entirely American row crop farmland, so strong sun, sandy soil, date palms, and circular centre pivot fields are all outside what the model has seen. The approach carries over fine. The trained weights would need real Saudi aerial imagery before anyone should trust them. Code, weights, sample images including at least one failure case, this write up, and a separate note on hours and AI use are all included in the repository.